

Minimax Algorithm

PROGRAM:

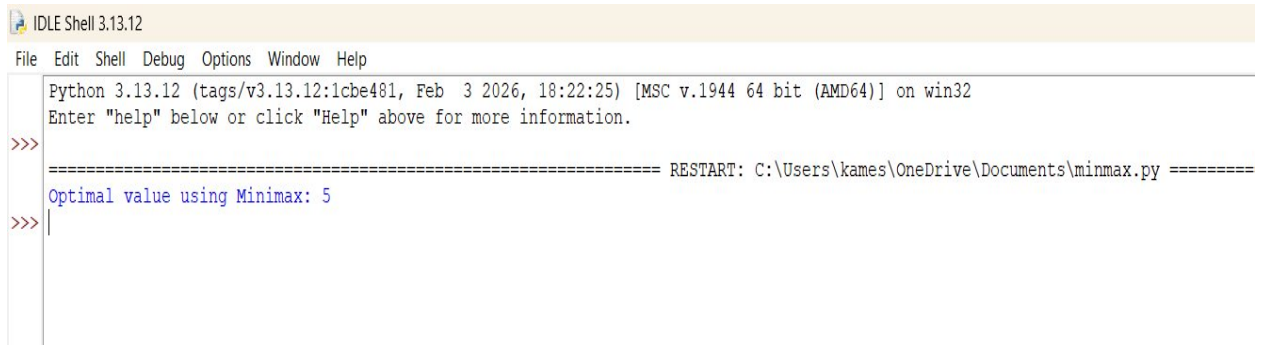
```
def minimax(depth, node_index, is_max, values, max_depth):

    if depth == max_depth:
        return values[node_index]

    if is_max:
        return max(
            minimax(depth + 1, node_index * 2, False, values, max_depth),
            minimax(depth + 1, node_index * 2 + 1, False, values, max_depth)
        )
    else:
        return min(
            minimax(depth + 1, node_index * 2, True, values, max_depth),
            minimax(depth + 1, node_index * 2 + 1, True, values, max_depth)
        )

values = [3, 5, 6, 9, 1, 2, 0, -1]
tree_depth = 3
result = minimax(0, 0, True, values, tree_depth)
print("Optimal value using Minimax:", result)
```

OUTPUT:



```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:\Users\kames\OneDrive\Documents\minmax.py =====
Optimal value using Minimax: 5
>>> |
```